31 MT
CRUDE OIL SUBSTITUTED SO FAR

93 MT
CARBON EMISSIONS AVOIDED

₹1.6 lakh cr.
ADDITIONAL FARM EARNINGS



CLAIM: Ants, bees swarm E20 as it contains sugar

GOVT RESPONSE: Fuel ethanol is distilled, contains no residual sugars, has denaturants



CLAIM: A litre of ethanol uses up 10,000 litres of water

GOVT RESPONSE: Ethanol plants use only 3-5 litres of processed water per litre, use zero-liquid-discharge systems



CLAIM: Ethanol absorbs water and ruins fuel tanks

GOVT RESPONSE: Modern vehicles have design safeguards to prevent this

components, which could raise maintenance and lifecycle costs over time.

The 2021 NITI Aayog roadmap itself acknowledged that oil marketing companies (OMCs) pass on changes in fuel costs due to ethanol blending to consumers. It warned that if ethanol became more expensive than petrol, ethanol-blended fuel could cost consumers more. That scenario has since materialised. According to the petroleum ministry, ethanol, which was cheaper than petrol in 2020-21, now costs more at the refinery stage. OMCs currently procure ethanol from distilleries at government-fixed prices of Rs 58-72 a litre, depending on the feedstock, while petrol's refinery transfer price stood at about Rs 70.5 a litre in early July. The ministry, therefore, argues that the programme is driven less by the promise of cheaper fuel than by energy security, lower crude imports, environmental goals and support for farmers (see *The E20 Debate*).

BEYOND THE FUEL TANK

For environmentalists, the debate goes beyond fuel cost or vehicle performance. The Centre for Science and Environment (CSE) argues that the aggressive push for ethanol risks placing pressure on water and food security. "Every litre of ethanol we burn in our cars costs us 10,000 litres of water from our farms," says CSE executive director Anumita Roychowdhury. "At a time when the Central Ground Water Board classifies large parts of Maharashtra, Uttar Pradesh and Punjab as over-exploited, diverting food crops into fuel tanks is not energy policy. It's water policy by the back door." She further warns that guaranteed demand for ethanol is encouraging farmers to shift from pulses, millets and oilseeds towards maize and sugarcane.

THE GOVERNMENT IS UNDERSTOOD TO BE INTERNALLY EVALUATING A POSSIBLE TRANSITION TO E21 AROUND 2027 AND E25 AROUND 2029

THE COST EQUATION

Vikram Gulati, executive vice president and country head of Toyota Kirloskar, says the West Asia crisis underlines India's "vulnerability" to imported crude. Ethanol blending, he argues, helped "mitigate the impact on the economy and the pockets of the consumers". But I.V. Rao, Distinguished Fellow at TERI and former head of R&D at Maruti Suzuki, says the consumer economics of E20 are more complicated than often portrayed. Since ethanol has lower energy content than petrol, he says, "a marginal drop in fuel efficiency is inevitable" because more fuel must be burnt to cover the same distance unless engines are specifically optimised for higher blends. He also points to the possible need for more E20-compatible fuel-system

Preparations for the next phase of ethanol blending are already under way. The Bureau of Indian Standards has notified fuel specifications for E22, E25, E27 and E30, while the Centre has begun consultations with vehicle manufacturers and other stakeholders on a staged transition beyond E20. In June, the government rolled out E85 for flex-fuel vehicles (FFVs) and exempted E22-E30 from central excise duty, though no rollout timeline has been announced for them. Officials are understood to be internally evaluating a shift to E21 around 2027 and E25 around 2029.

Internationally, there is no single template. Brazil mandates E30 as its standard blend but has a predominantly FFV fleet capable of running on blends of up to E100. The US, by contrast, has retained E10 as its standard fuel while expanding access to E15 and supporting millions of FFVs that can run on up to E85. Japan has virtually no ethanol blending but plans to introduce E10 by 2030 and E20 by 2040.

The petroleum ministry seeks to dissipate such concerns by saying modern ethanol plants consume only three to five litres of processed water to produce a litre of ethanol and increasingly operate with zero-liquid-discharge systems. It also argues that only surplus rice, identified by the Department of Food and Public Distribution after meeting national food security requirements, is diverted for ethanol production.

India's ethanol experiment is no longer confined to government roadmaps or lab reports. Motorists measure it every day in kilometres per litre; policymakers, in barrels of crude not imported. Whether the two align will determine E20's success. ■